

Title: Assignment 1 Digital Forensics

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Case Information

Case ID / Reference number	1
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Examiner(s)	Athallah Rajendra Wibisono
Requesting party (e.g., course / lecturer)	Digital Forensics Course / Hudan Studiawan, S.Kom., M.Kom., Ph.D.
Case type (e.g., mobile forensics, disk forensics, network forensics)	Disk forensics
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Examiner(s)	Muhammad Rengga Putra Kuncoro
Requesting party (e.g., course / lecturer)	Digital Forensics Course / Hudan Studiawan, S.Kom., M.Kom., Ph.D.
Case type (e.g., mobile forensics, disk forensics, network forensics)	Image forensics
Date and location of examination	October 5th-6th 2026, Online

Case ID 1: Disk Forensics

Introduction

Given a USB flash drive, we attempt to recover deleted files in that drive. To do so, we save any file to that drive (in our case, it would be an image of a cat) and permanently delete it (using Shift + Delete). Then, we will recover said file with a tool, namely TestDisk.

Scope and Objectives

We aim to recover a deleted file from a USB flash drive. Due to storage limitations and the big size of our USB drive (115GB), we did not create a disk image and perform an investigation on the aforementioned image. Instead, we perform the investigation on the live, plugged-in USB. However, we are aware that this decision deviates from forensics best practices, hence we will provide the steps on what we would do should we have no storage limitations. To mimic forensics best practices, we mount the USB as read-only to prevent altering the digital evidence.

Tools Used

No.	Tool	Version	Source / URL	Purpose
1	TestDisk	7.3	https://www.cgsecurity.org/wiki/TestDisk_Download	Creating forensic/disk image & recovering deleted file

Description of Digital Evidence

A forensic disk image like a raw **.dd** file would be a good digital evidence to perform disk investigation and recovery. However, since we do not investigate from a disk image (due to storage limitations), we perform recovery directly on a live USB drive. Below is the specifications of the drive:

```
DISKPART> detail disk

Kingston DataTraveler 3.0 USB Device
Disk ID: {C681B086-C42D-435C-95E6-633D6C197709}
Type   : USB
Status : Online
Path   : 0
Target : 0
LUN ID : 0
Location Path : UNAVAILABLE
Current Read-only State : Yes
Read-only  : Yes
Boot Disk  : No
Pagefile Disk : No
Hibernation File Disk : No
Crashdump Disk : No
Clustered Disk : No

Volume ###  Ltr  Label      Fs      Type        Size      Status      Info
-----
Volume 4     E             exFAT    Removable   115 GB    Healthy
```

Forensic Process Performed

1. Preparation

Before we begin our investigation, we download the tool(s) necessary. For this assignment, only TestDisk is needed, so we download it from its official website https://www.cgsecurity.org/wiki/TestDisk_Download

Beta: TestDisk & PhotoRec 7.3-WIP, Data Recovery

For more information, read the [7.3 release notes](#) and the [git history](#).

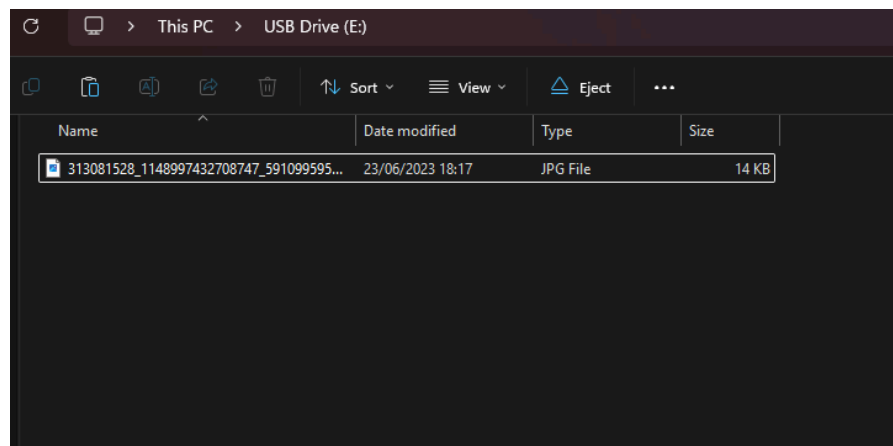
Select your operating system to download the latest version of TestDisk & PhotoRec data recovery tools.

- [Windows 64-bit](#)
- [Windows 32-bit](#), minimum requirement: Windows Vista, Windows Server 2008 and above.
- [Mac OS X Intel 64-bit](#) macOS >= 10.15 (Note: Mac M1 can run this version.)
- [Linux x86_64](#), kernel 2.6.18 or later
- [Dos/Win95/Win98](#)
- [TestDisk & PhotoRec Source code](#)
- Files hashes [sha256](#) [sha512](#) (WIP files may change over time).

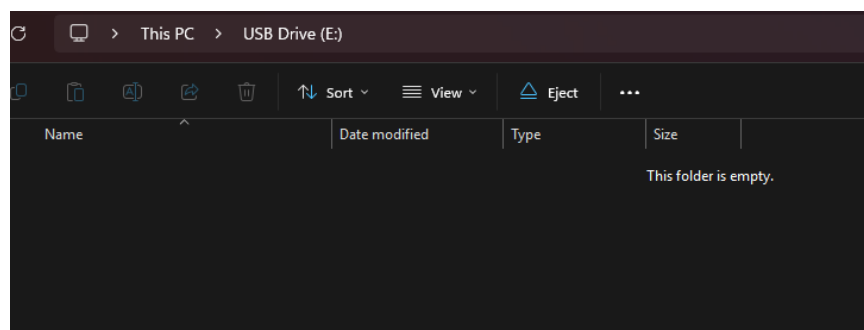


After downloading, we simply unzipped the file and ensured `testdisk_win.exe` exists, since that is the binary that we will use to complete this assignment.

Then, we prepare the scenario by creating a file inside the USB flash drive and permanently deleting it. We added a simple image file of a cat.



Then, we delete it.



2. Collection (Identification, Acquisition, and Preservation)

Ideally, we would create a forensic disk image, such as a raw **.dd** file. To do that, we run the TestDisk binary (for windows, it's the **testdisk_win.exe** file), then select the disk (and partition) we want to get its image.

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 480 GB / 447 GiB - KINGSTON SA400S37480G
Disk \\.\PhysicalDrive1 - 512 GB / 476 GiB - NVMe KINGSTON OM8PDP3512B-AA1
>Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - Kingston DataTraveler 3.0

>[Proceed] [Quit]

Note: Serial number EE07DB5156E9
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - Kingston DataTraveler 3.0

Please select the partition table type, press Enter when done.
[Intel] Intel/PC partition
>[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax] Humax partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] Xbox partition
[Return] Return to disk selection

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

Then, we go to Advanced → Image Creation and select where we want to create the disk image.

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - Kingston DataTraveler 3.0
CHS 15053 255 63 - sector size=512

[Analyse] Analyse current partition structure and search for lost partitions
>[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - CHS 15053 255 63

Partition              Start          End      Size in sectors
> 1 P MS Data           2048    241825791  241823744 [Basic data partition]

[ Type ] [ Boot ] [ List ] [Undelete] >[Image Creation] [ Quit ]
```

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select where to store the file image.dd (123813 MB), an image of the
partition
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory C:\Users\Acer\Desktop\its\semester 7
>drwxr-x--- 197609 197121      0 5-Oct-2026 11:18 .
drwxr-x--- 197609 197121      0 18-Sep-2026 14:19 ..
drwxr-x--- 197609 197121      0 5-Oct-2026 10:56 testdisk-7.3-WIP.win64
-rwxr-x--- 197609 197121 215693 18-Sep-2026 14:19 Rubrik penilaian tugas 3 bindo.pdf
```

Afterwards, a disk image creation process should start. TestDisk will create an exact copy of the whole disk to this file, so the resulting image will be as big as the drive.

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - Kingston DataTraveler 3.0
1 P MS Data           2048    241825791  241823744 [Basic data partition]
0.33 % =>

Disk images are mainly used
- for forensic purposes
- or to deal with media with bad sectors

To use TestDisk or PhotoRec with this disk image, start a Terminal and run
  testdisk image.dd
or photorec image.dd
```

Finally, we would run `testdisk image.dd` to run analysis on the disk image. However, since we analyze the live USB directly, we instead mount/plug in the USB and make it read-only to mimic a write-blocker (to prevent tampering with the digital evidence), although we are aware that this is a weaker software-based protection compared to an actual write-blocker.

To do that, we run `diskpart` in a terminal, then select the appropriate disk (in our case, its disk 2).

```
DISKPART> list disk

Disk ###  Status       Size       Free       Dyn  Gpt
-----  -
Disk 0    Online      447 GB     117 GB
Disk 1    Online      476 GB      0 B
Disk 2    Online      115 GB      0 B

DISKPART> select disk 2

Disk 2 is now the selected disk.
```

To make it read-only, we simply run `attributes disk set readonly`.

```
DISKPART> attributes disk set readonly

Disk attributes set successfully.
```

Running `detail disk` will show that the disk is now indeed read-only.

```
DISKPART> detail disk

Kingston DataTraveler 3.0 USB Device
Disk ID: {C681B086-C42D-435C-95E6-633D6C197709}
Type : USB
Status : Online
Path : 0
Target : 0
LUN ID : 0
Location Path : INAVAILABLE
Current Read-only State : Yes
Read-only : Yes
Boot Disk : No
Pagefile Disk : No
Hibernation File Disk : No
Crashdump Disk : No
Clustered Disk : No
```

3. Examination

To start the recovery, run the TestDisk binary and select the appropriate disk. Then, select “Analyze”

```

TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - Kingston DataTraveler 3.0
CHS 15053 255 63 - sector size=512

> [ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.

```

Find the partition. When the correct partition has been found, select it and press “P” to list all its files, including the deleted ones.

```

TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 123 GB / 115 GiB - CHS 15053 255 63
Partition      Start      End      Size in sectors
SP MS Data      2048      241825791  241823744

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
                P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
exFAT, blocksize=131072, 123 GB / 115 GiB

```

```

C:\Users\Acer\Desktop\its\semester_7\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P MS Data 2048 241825791 241823744
Directory /
>drwxr-xr-x 0 0 131072 5-Oct-2026 10:51 System Volume Information
-rwxr-xr-x 0 0 13879 23-Jun-2023 18:17 313081528_1148997432708747_5910995952251052677_n.jpg

Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, '.' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file

```

Then, select the file we want to recover, and decide where we want to recover it. Since we are performing this on a live USB, it is imperative that the destination is in a different partition/disk, since copying to the same disk can risk overwriting it.

```

C:\Users\Acer\Desktop\its\semester_7\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select a destination where /
/313081528_1148997432708747_5910995952251052677_n.jpg will be copied.
Keys: Arrow keys to select another directory
C when the destination is correct
Q to quit
Directory C:\Users\Acer\Desktop\its\semester_7
>drwxr-x-- 197609 197121 0 5-Oct-2026 11:10 .
drwxr-x-- 197609 197121 0 18-Sep-2026 14:19 ..
drwxr-x-- 197609 197121 0 5-Oct-2026 10:56 testdisk-7.3-WIP.win64
-rwxr-x-- 197609 197121 215693 18-Sep-2026 14:19 Rubrik penilaian tugas 3 bindo.pdf

```

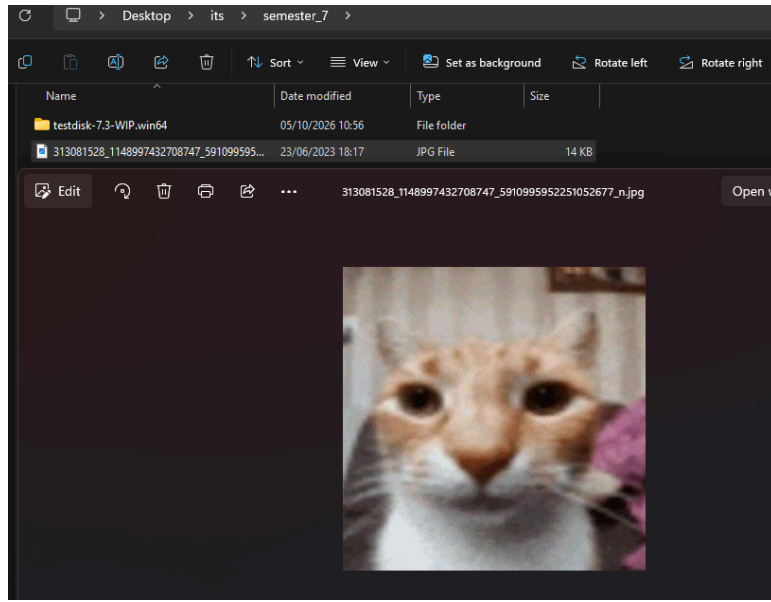
```

P MS Data 2048
Directory /313081528_1148997432708747_
Copy done! 1 ok, 0 failed
drwxr-xr-x 0 0 131072 5-0
>-rwxr-xr-x 0 0 13879 23-J

```

4. Analysis

To see the result, we can simply open the recovered file in our targeted directory.



5. Reporting

The result is that deleted files can be recovered back in a disk, as long as the drive is not used much. Another person can reproduce it from the forensic disk image that we (would) create via TestDisk.

Findings

Finding ID	Artifact / data	Source (item and path)	Description	Significance
F-01	JPG file	313081528_1148997432708747_5910995952251052677_n.jpg	A deleted image file of a cat	Deleted file can be recovered successfully

Limitations and Assumptions

The recovery is possible because the disk is not used between deletion and recovery process. When a file is deleted, the device marks that file's former area as "writable", meaning that frequent usage or writing to the disk may risk overwriting the old data, making chances of recovery smaller.

Conclusion

Using TestDisk, we can successfully recover deleted files in a disk, such as a USB flash drive or from a forensic disk image. For the former, however, the drive should not be used as usage can risk making file recovery harder.

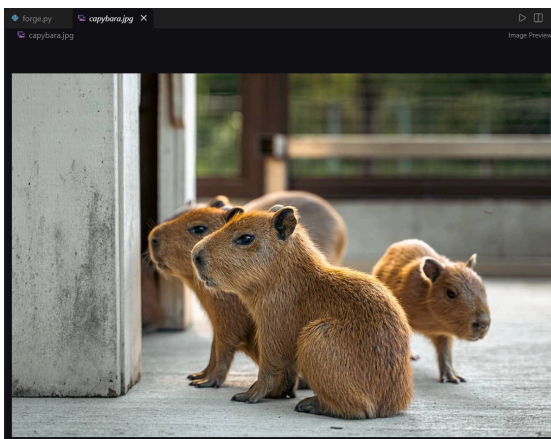
Case ID 2: Image Forensics

1. Preparation

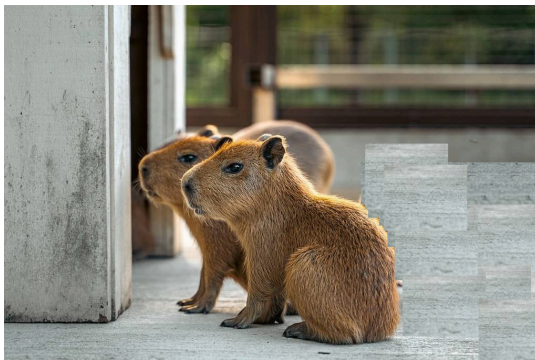
Before we begin our investigation, we need to install the tools we need in order to do Image Forensics. With the goal to test and analyze image copy-move forgery, we need to install Sherlock

<https://github.com/GuidoBartoli/sherlock>

We then prepare the image, here we have an image of capybaras



To apply copy-move forgery we use Canva to try and hide the third capybara by copying and moving certain parts of the image:



We also create a script to generate the “ground truth” image by masking the edited parts to use as validation.

```

mask.py > ...
1  import cv2
2  import numpy as np
3
4  def make_mask(orig_path, tampered_path, out_path, thresh=25):
5      """Write a binary mask of pixels that differ between original and tampered."""
6      a = cv2.imread(orig_path)
7      b = cv2.imread(tampered_path)
8      diff = cv2.absdiff(a, b).max(axis=2)
9      mask = (diff > thresh).astype(np.uint8) * 255
10     mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, np.ones((7, 7), np.uint8))
11     mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, np.ones((3, 3), np.uint8))
12     cv2.imwrite(out_path, mask)
13
14     make_mask("copybara.jpg", "copybara_canva.png", "copybara_canva_mask.png")

```

2. Collection (Identification, Acquisition, and Preservation)

We received one image, copybara_canva.png (PNG, 876,315 bytes), with a claim that it was manipulated. No original, capture device or editing history was provided. We logged the supplier, the delivery method, the date, the filesystem timestamps and whether a Zone.Identifier stream was present.

The file is a lossless PNG, so JPEG-specific tools (ELA, JPEG Ghosts and quantization tables) have little to work with. The Copy-Move, Noise and Wavelet tools still apply. Its metadata sits in PNG chunks (IHDR, tEXt, iCCP, pHYs, and similar), which can show how the file was exported.

00 00 00 08 00 04	MM.*.....	TIFF header IFD0 entries
00 00 00 01 00 00 00 3e	>	IFD0-00 XResolution
00 00 00 01 00 00 00 46	F	IFD0-01 YResolution
00 00 00 01 00 02 00 00	.(.....	IFD0-02 ResolutionUnit
00 00 00 01 00 01 00 00		IFD0-03 YCbCrPositioning
00 00 00 00 00 00 00 00		Next IFD
00 01 00 00 00 60 00 00 00 01 70 00	.`.....`....p.	XResolution value YResolution value
3c 3f 78 70 61 63 6b 65 74 20 62 65	<?xpacket be	[unknown data]
27 ef bb bf 27 20 69 64 3d 27 57 35	gin='...' id='W5	

	Group	Description	Value
1	File	ZoneIdentifier	Exists
2		FileCreateDate	2026:10:06 16:51:39+07:00
3		ExifByteOrder	MM
4	PNG	ImageWidth	1024
5		ImageHeight	683
6		BitDepth	8
7		ColorType	2
8		PixelsPerUnitX	3780
9		PixelsPerUnitY	3780
10		PixelUnits	1
11	XMP	AdsCreated	2026-10-06
12		AdsData	{"doc":"DAHKO0IMbvQ","user":"UAF6BPpHt3w","brand":"BAF6BGwKERY"}
13		AdsExtId	59c95eb8-2a9b-45ca-b29d-c97ebd5cecf2
14		AdsFbId	525265914179580
15		AdsTouchType	2
16		Title	Untitled design - 1
17		Author	tend5 天道 (tendonintendo)
18		CreatorTool	Canva (Renderer) doc=DAHKO0IMbvQ user=UAF6BPpHt3w brand=BAF6BGwKERY
19	EXIF	XResolution	96
20		YResolution	96

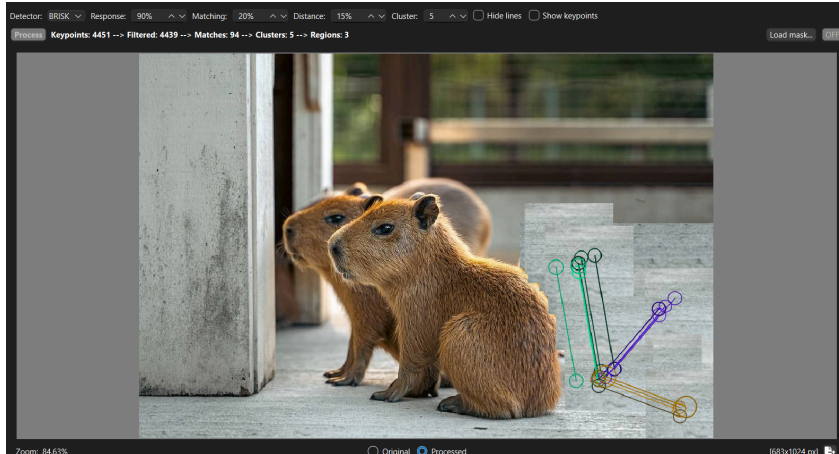
Sherloq's Header Structure shows an embedded EXIF/TIFF block with a single IFD0 containing four entries: XResolution, YResolution, ResolutionUnit and YCbCrPositioning, followed by the values of the two resolution entries. The next-IFD pointer is empty, so there is no Exif sub-IFD, GPS IFD or thumbnail. The byte order is big-endian (MM). No Make, Model, DateTime, Software or lens tags are present, so there is no camera or capture evidence.

The ExifTool export adds the PNG and XMP details. The image is 1024x683, 8-bit RGB (color type 2) at 3780 pixels per meter, about 96 DPI, which is the screen default and not a capture value. The XMP block contains Canva-specific fields: the title "Untitled design - 1", an AdsCreated date of 2026-10-06, an AdsData field holding document, user and brand identifiers, and ad/tracking IDs. The file also has a Zone.Identifier stream, which means it was downloaded.

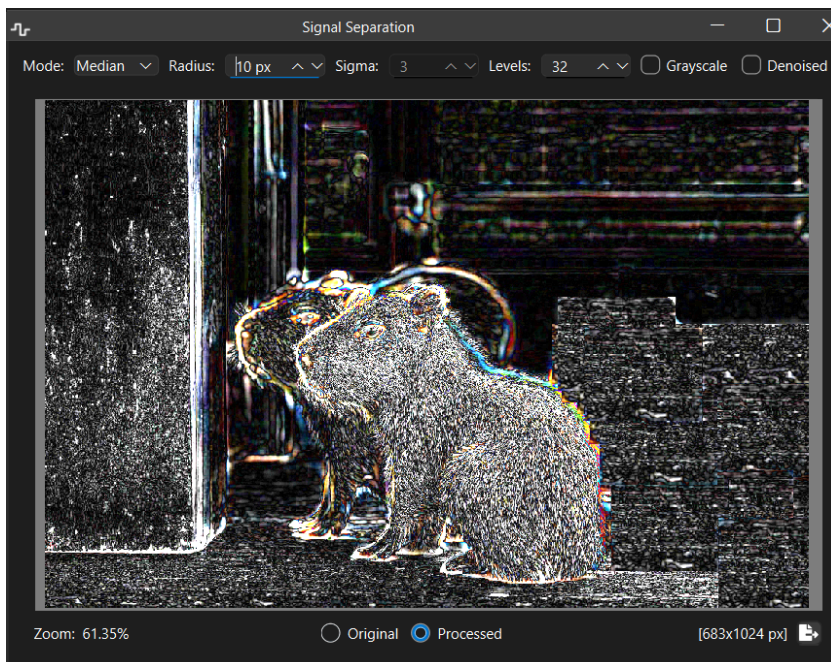
Taken together, the metadata shows the image was exported from Canva on 2026-10-06 and then downloaded. It contains no capture information, and it can't show which edits were made. That requires pixel-level analysis.

3. Examination

Using the Tampering tool in Sherloq to detect Copy-Move Forgery, we were able to detect the duplicated regions as clusters using the BRISK detector



To clearly visualize the edited regions, we are able to see the harsh edges by implementing a noise filter on the image, specifically signal separation.



Due to the poorly edited image, causing odd-looking lighting, we can see the edges even clearer using an echo edge filter



4. Analysis

It can be seen that we were able to highlight and outline the edited parts of the image using the tools available in Sherloq, even being able to detect the clusters using the Copy-Move Forger tool. This can be validated using the previously generated mask.



5. Reporting

The result is that the image was exported from Canva and contains duplicated regions consistent with copy-move forgery. Another person can reproduce it by hashing the evidence file, opening the working copy in Sherloq and applying the same tools and settings listed below.

Finding ID	Artifact / data	Source (item and path)	Description	Significance
------------	-----------------	------------------------	-------------	--------------

F-01	PNG file	copybara_canva.png (876,315 bytes, SHA-256: [hash])	Suspect image of copybaras, 1024x683, 8-bit RGB, received with a claim of manipulation. No original, device or editing history was supplied.	Identifies the evidence. The hash confirms its integrity.
F-02	XMP metadata	copybara_canva.png	Title "Untitled design - 1", AdsCreated 2026-10-06, Canva document/user/brand IDs and ad/tracking IDs.	The file was exported from Canva, not from a camera. The IDs could identify the creating account if Canva is asked.
F-03	EXIF/TIFF header	copybara_canva.png (Sherloq Header Structure)	IFD0 holds only XResolution, YResolution, ResolutionUnit and YCbCrPositioning. Big-endian (MM). No Make, Model, DateTime, Software, GPS or thumbnail. About 96 DPI (3780 px/m).	No capture evidence exists. The metadata can't show which edits were made.
F-04	Zone.Identifier	copybara_canva.png	Stream present.	The file was downloaded rather than created on the analysis machine.
F-05	Duplicated regions	copybara_canva.png (Sherloq Tampering > Copy-Move Forgery, BRISK detector)	Matched keypoint clusters between duplicated regions of the image, in the area where the third copybara was hidden. [Screenshot]	Indicates regions were copied and moved within the image, consistent with copy-move forgery.
F-06	Noise inconsistency	copybara_canva.png (Sherloq Signal Separation)	The noise filter shows harsh edges around the edited regions. [Screenshot]	The edited regions differ from the surrounding

				image, which supports F-05.
F-07	Edge/lighting artifacts	copybara_canva.png (Sherloq Echo Edge Filter)	Edges of the edited regions stand out more clearly because of the odd-looking lighting from the poor edit. [Screenshot]	A second, independent tool outlines the same regions, which supports F-05 and F-06.

Limitations and Assumptions

- No original image was available, so the analysis only looks for internal inconsistencies.
- The file is a lossless PNG, so JPEG-based tools (ELA and Double JPEG) were not applicable.
- Canva re-rasterizes on export, which can remove noise and resampling traces. A negative result doesn't prove the image is authentic.
- Natural repeating textures (fur, grass) can cause false matches, so each finding was cross-checked with a second tool.
- The metadata could have been altered or stripped, so it supports provenance only as a lead.
- Hashes were verified before and after each session. The read-only attribute is a weaker substitute for a hardware write-blocker.

Conclusion

Using Sherloq, we are able to find modifications and abnormalities that exist in images and decide whether or not they have been tampered with.